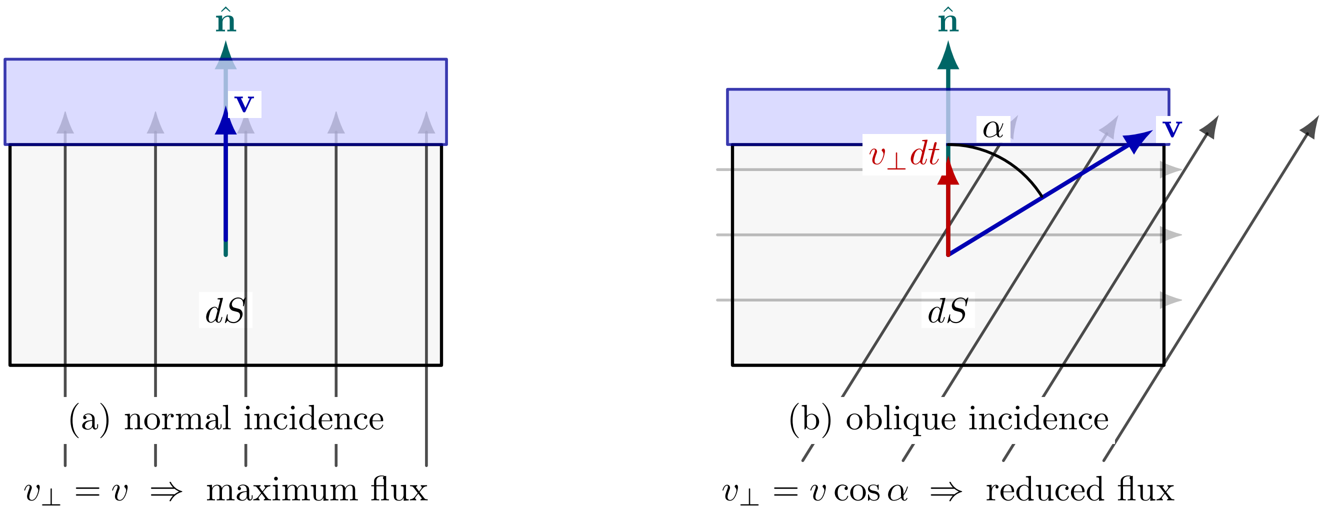
Concept of Flux and the Divergence Theorem (Leading to Electric Flux and Gauss Theorem)

1) Intuitive meaning of flux (why $\hat{\mathbf{n}}$ and dot product?)

What flux is. Think of a velocity field $\mathbf{v}(\mathbf{r})$ describing water flow. Place a *thin net / membrane* (a small surface patch) of area dS in the flow. *Flux* is simply the **volume of water per second** that passes *through* that patch.



$$d\Phi = \frac{dV}{dt} = (\mathbf{v}\cdot\hat{\mathbf{n}})\,dS = v\cos\alpha\,dS$$

Figure 1: Why the dot product appears in flux: only the perpendicular component of the flow crosses the surface patch.

If the patch is part of a larger surface S, the **total flux** is the total "through–flow" across all its tiny patches.

Step 1: Only the perpendicular component can cross the surface

Take a tiny flat patch. Decompose the flow vector into two parts:

$$\mathbf{v} = \mathbf{v}_\perp + \mathbf{v}_\parallel, \qquad \mathbf{v}_\perp \parallel \hat{\mathbf{n}}, \quad \mathbf{v}_\parallel \perp \hat{\mathbf{n}}.$$

Key physical point:

- $\mathbf{v}_{\parallel}$ is *tangential* to the surface. It makes water *slide along* the surface, but it does *not* push water through it.
- $\mathbf{v}_{\perp}$ points *across* the surface. This part alone determines how much water crosses the patch.

Hence, the "through-flow" depends only on $v_{\perp}$.

Step 2: A geometric derivation of $d\Phi = v_{\perp}\, dS$

During a short time dt, the water that crosses the patch forms a thin *slab* of thickness

$$\text{thickness} = v_{\perp}\, dt.$$

So the volume that passes through the patch in time dt is

$$dV = (\text{area}) \times (\text{thickness}) = dS\,(v_{\perp}\, dt).$$

Therefore the **volume flow rate** through the patch (volume per second) is

$$\frac{dV}{dt} = v_{\perp}\, dS.$$

We call this **differential flux**:

$$d\Phi \;\equiv\; v_{\perp}\, dS.$$

Step 3: Why the dot product appears

Let α be the angle between $\mathbf{v}$ and the outward unit normal $\hat{\mathbf{n}}$. Then the perpendicular component is

$$v_{\perp} = \|\mathbf{v}\| \cos\alpha.$$

But $\|\mathbf{v}\| \cos\alpha$ is exactly the dot product:

$$\mathbf{v} \cdot \hat{\mathbf{n}} = \|\mathbf{v}\|\, \|\hat{\mathbf{n}}\| \cos\alpha = \|\mathbf{v}\| \cos\alpha = v_{\perp}.$$

So the differential flux becomes the compact expression

$$d\Phi = (\mathbf{v} \cdot \hat{\mathbf{n}})\, dS.$$

Interpretation:

- If $\mathbf{v}$ points outward, $\mathbf{v} \cdot \hat{\mathbf{n}} > 0$ (positive flux).
- If $\mathbf{v}$ points inward, $\mathbf{v} \cdot \hat{\mathbf{n}} < 0$ (negative flux).
- If $\mathbf{v}$ is tangent to the surface, $\mathbf{v} \cdot \hat{\mathbf{n}} = 0$ (no crossing).

Step 4: Total flux through a surface

Adding (integrating) all patches over S:

$$\Phi = \iint_S \mathbf{v} \cdot \hat{\mathbf{n}}\, dS = \iint_S \mathbf{v} \cdot d\mathbf{S}, \quad \text{where } d\mathbf{S} = \hat{\mathbf{n}}\, dS.$$

The same definition applies in electrostatics by replacing $\mathbf{v}$ with $\mathbf{E}$:

$$\Phi_E = \iint_S \mathbf{E} \cdot \hat{\mathbf{n}}\, dS.$$

2) An Intuitive (Local-to-Global) Proof of the Divergence Theorem

Idea

Take a tiny rectangular box $\Delta V = \Delta x\, \Delta y\, \Delta z$ around a point $P(x, y, z)$ inside a closed surface S. Compute the net outward flux through the tiny box. This gives $(\nabla \cdot \mathbf{F})\Delta V$. Then fill the whole volume with many such boxes: internal-face flux cancels, leaving only the flux through S.

Step 1: Flux through a tiny box around $P(x, y, z)$

Let $\mathbf{F}(x, y, z) = F_1\, \hat{\mathbf{i}} + F_2\, \hat{\mathbf{j}} + F_3\, \hat{\mathbf{k}}$.

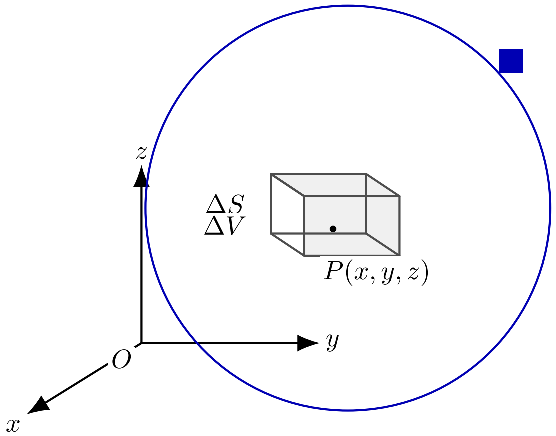


Figure 2: Zoom into an infinitesimal box inside a closed surface S. We first compute flux through the box; then sum over all boxes in the volume.

(a) Contribution from the two x-faces

Consider the two faces perpendicular to the x-axis:

$$\text{Left face at } x: \quad \hat{\mathbf{n}} = -\hat{\mathbf{i}}, \quad dS = \Delta y\, \Delta z,$$

$$\text{Right face at } x + \Delta x: \quad \hat{\mathbf{n}} = +\hat{\mathbf{i}}, \quad dS = \Delta y\, \Delta z.$$

Flux through the two x-faces:

$$\Phi_x = \underbrace{\mathbf{F}(x+\Delta x, y, z)\cdot(+\hat{\mathbf{i}})\,\Delta y\,\Delta z}_{\text{right face}} + \underbrace{\mathbf{F}(x, y, z)\cdot(-\hat{\mathbf{i}})\,\Delta y\,\Delta z}_{\text{left face}}.$$

Since $\mathbf{F}\cdot\hat{\mathbf{i}} = F_1$,

$$\Phi_x = F_1(x+\Delta x, y, z)\,\Delta y\,\Delta z - F_1(x, y, z)\,\Delta y\,\Delta z = \big[F_1(x+\Delta x, y, z) - F_1(x, y, z)\big]\Delta y\,\Delta z.$$

Rewrite as a difference quotient:

$$\Phi_x = \left[\frac{F_1(x+\Delta x, y, z) - F_1(x, y, z)}{\Delta x}\right]\Delta x\,\Delta y\,\Delta z.$$

In the limit $\Delta x \to 0$,

$$\boxed{\Phi_x \;\to\; \frac{\partial F_1}{\partial x}\,\Delta V.}$$

(b) Similarly for y- and z-faces

By the same argument,

$$\boxed{\Phi_y \to \frac{\partial F_2}{\partial y}\,\Delta V, \qquad \Phi_z \to \frac{\partial F_3}{\partial z}\,\Delta V.}$$

(c) Total flux through the tiny closed box

Add all six faces:

$$\Phi_{\text{box}} = \Phi_x + \Phi_y + \Phi_z \to \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}\right)\Delta V = (\nabla\cdot\mathbf{F})\,\Delta V.$$

Hence the key local statement:

$$\boxed{\oiint_{\partial(\Delta V)} \mathbf{F}\cdot d\mathbf{S} \;=\; (\nabla\cdot\mathbf{F})\,\Delta V \quad \text{(in the limit of an infinitesimal box).}}$$

Step 2: From a tiny box to the entire closed surface

Now fill the entire volume V enclosed by S with many tiny boxes ΔV_k.

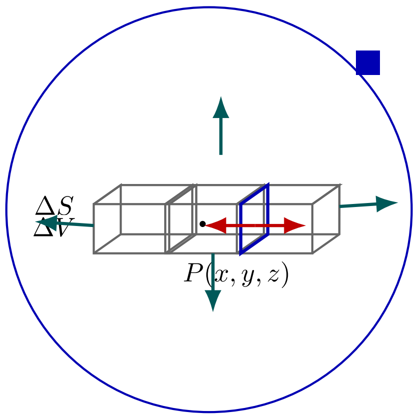


Figure 3: When the volume is partitioned into small boxes, flux through shared internal faces cancels pairwise (opposite outward normals). Only flux through the outer boundary $S = \partial V$ remains.

- Sum the flux of each box:

$$\sum_k \oiint_{\partial(\Delta V_k)} \mathbf{F} \cdot d\mathbf{S} = \sum_k (\nabla \cdot \mathbf{F})\, \Delta V_k.$$

- Flux through *internal faces cancels*: every internal face is shared by two neighboring boxes, and the outward normals are opposite, so the flux contributions are equal and opposite.

- Only the outer boundary survives, so the left side collapses to the flux through S:

$$\sum_k \oiint_{\partial(\Delta V_k)} \mathbf{F} \cdot d\mathbf{S} = \oiint_S \mathbf{F} \cdot d\mathbf{S}.$$

- Taking the limit of infinitely fine boxes converts the right-hand sum to a volume integral:

$$\sum_k (\nabla \cdot \mathbf{F})\, \Delta V_k \to \iiint_V (\nabla \cdot \mathbf{F})\, dV.$$

Therefore,

$$\boxed{\oiint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (\nabla \cdot \mathbf{F})\, dV.}$$

This is the **divergence theorem**.

Intuitive meaning

Intuitive Interpretation

- $\nabla \cdot \mathbf{F}$ measures the *net outflow generated locally* per unit volume.
- Integrating $\nabla \cdot \mathbf{F}$ over the volume adds up all local sources/sinks inside.
- The surface flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ is the *total* outflow crossing the boundary.
- Equality says: *Total outflow through the boundary = total source strength inside.*

3) Electric Flux and First Maxwell's Equation: Gauss-Law

Electric flux: definition

For an oriented surface S with outward unit normal $\hat{\mathbf{n}}$, the **electric flux** is

$$\boxed{\Phi_E = \iint_S \mathbf{E} \cdot \hat{\mathbf{n}}\, dS.}$$

If S is a *closed* surface, we write

$$\boxed{\Phi_E = \oiint_S \mathbf{E} \cdot d\mathbf{S} = \oiint_S \mathbf{E} \cdot \hat{\mathbf{n}}\, dS.}$$

Case 1: point charge Q at the origin; flux through a sphere of radius r

The electric field due to a point charge at the origin is

$$\boxed{\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r^2}\, \hat{\mathbf{r}}.}$$

Consider the closed surface S as a sphere of radius r centered at the origin. On this sphere:

$$\hat{\mathbf{n}} = \hat{\mathbf{r}} \qquad \text{and} \qquad \mathbf{E} \parallel \hat{\mathbf{n}}.$$

Therefore,

$$\mathbf{E} \cdot \hat{\mathbf{n}} = E_r = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r^2}.$$

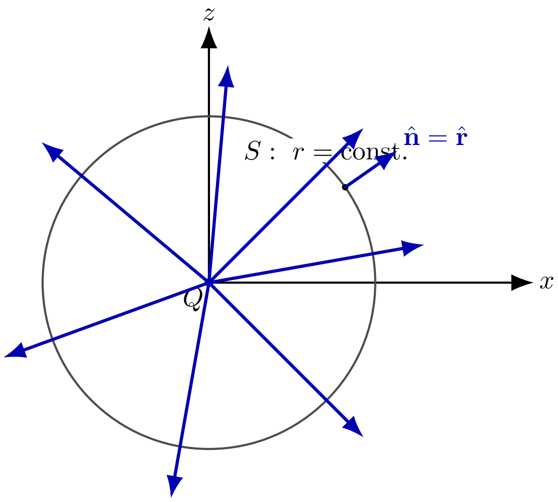
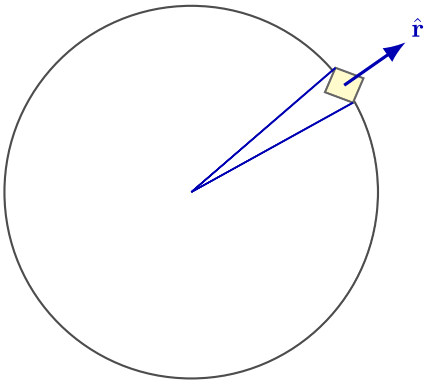


Figure 4: For a sphere centered at the point charge, $\mathbf{E}$ is radial everywhere and parallel to the outward normal $\hat{\mathbf{n}} = \hat{\mathbf{r}}$.

Surface element on a sphere

Using spherical coordinates (r, θ, ϕ) with θ the polar angle from $+z$ and ϕ the azimuth:

$$\boxed{dS = r^2 \sin\theta \, d\theta \, d\phi.}$$



$dS = r^2 \sin\theta \, d\theta \, d\phi$

Figure 5: A small surface element dS on a sphere. Extending rays from the origin to the patch outlines a narrow cone corresponding to the solid angle subtended by the patch.

Flux calculation

$$\Phi_E = \oiint_S \mathbf{E} \cdot \hat{\mathbf{n}} \, dS = \oiint_S E_r \, dS.$$

Substitute $E_r = \dfrac{1}{4\pi\varepsilon_0}\dfrac{Q}{r^2}$ and $dS = r^2 \sin\theta\, d\theta\, d\phi$:

$$\begin{aligned}
\Phi_E &= \int_0^{2\pi}\int_0^{\pi}\left(\frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\right)\left(r^2\sin\theta\, d\theta\, d\phi\right)\\
&= \frac{Q}{4\pi\varepsilon_0}\int_0^{2\pi}\int_0^{\pi}\sin\theta\, d\theta\, d\phi\\
&= \frac{Q}{4\pi\varepsilon_0}\left(\int_0^{2\pi} d\phi\right)\left(\int_0^{\pi}\sin\theta\, d\theta\right)\\
&= \frac{Q}{4\pi\varepsilon_0}(2\pi)\Big[-\cos\theta\Big]_0^{\pi}\\
&= \frac{Q}{4\pi\varepsilon_0}(2\pi)(2)\\
&= \boxed{\frac{Q}{\varepsilon_0}}.
\end{aligned}$$

Key observation (why radius does not matter). On a sphere, $|\mathbf{E}| \propto \dfrac{1}{r^2}$ but $dS \propto r^2$; the factors cancel, so the total flux depends only on the enclosed charge, not on the sphere radius.

Key Takeaways

- Electric flux through a (closed) surface is $\Phi_E = \oiint_S \mathbf{E}\cdot d\mathbf{S}$.
- For a point charge at the center of a sphere, $\mathbf{E} \parallel \hat{\mathbf{n}}$ everywhere on the sphere.
- On a sphere, $E \propto 1/r^2$ and $dS \propto r^2$ cancel $\Rightarrow$ total flux is independent of r.
- Result: $\oiint_S \mathbf{E}\cdot d\mathbf{S} = \dfrac{Q}{\varepsilon_0}$ (For a point charge at center of the spherical surface S).

Case 2. Flux of a Point Charge Inside a Sphere (Charge Not at the Center)

Let a point charge Q be located at $\mathbf{r}_Q$ (not at the sphere center). Let S be a sphere (or any closed surface) that *encloses* the charge.
The electric field at a surface point $\mathbf{r} \in S$ is

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\frac{Q(\mathbf{r}-\mathbf{r}_Q)}{\|\mathbf{r}-\mathbf{r}_Q\|^3} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R^2}\hat{\mathbf{R}}, \quad \mathbf{R} = \mathbf{r}-\mathbf{r}_Q,\ R = \|\mathbf{R}\|.$$

Key geometric idea: relate dS to a solid angle $d\Omega$

Pick a small surface patch dS around a point on S. Let $\hat{\mathbf{n}}$ be the outward normal to S at that patch. Let α be the angle between $\hat{\mathbf{R}}$ (direction from charge to patch) and $\hat{\mathbf{n}}$.
Then

$$\mathbf{E}\cdot d\mathbf{S} = \mathbf{E}\cdot(\hat{\mathbf{n}}\,dS) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R^2}\,(\hat{\mathbf{R}}\cdot\hat{\mathbf{n}})\,dS = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R^2}\,(\cos\alpha)\,dS.$$

Define the *projected area*:

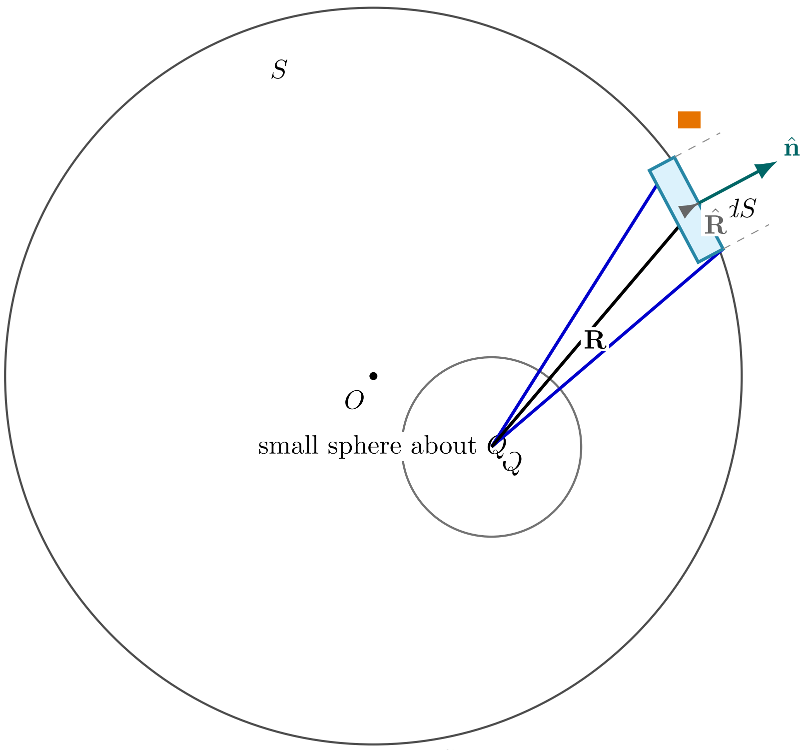
$$dS_\perp \;=\; dS\cos\alpha.$$

Hence the solid angle:

$$\boxed{d\Omega \;=\; \frac{dS_\perp}{R^2} \;=\; \frac{dS\cos\alpha}{R^2}.}$$

Substitute into the flux element:

$$\boxed{\mathbf{E}\cdot d\mathbf{S} = \frac{1}{4\pi\varepsilon_0}\,Q\,d\Omega.}$$



$$d\Omega = \frac{dS\cos\alpha}{R^2}$$

For the field of a point charge,

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R^2}\hat{\mathbf{R}}, \quad R = \|\mathbf{r}-\mathbf{r}_Q\|.$$

The flux through the small patch is

$$\mathbf{E} \cdot d\mathbf{S} = \frac{1}{4\pi\varepsilon_0} \frac{Q}{R^2} (\hat{\mathbf{R}} \cdot \hat{\mathbf{n}})\, dS = \frac{Q}{4\pi\varepsilon_0} \frac{dS \cos\alpha}{R^2} = \frac{Q}{4\pi\varepsilon_0} d\Omega.$$

Integrating over any closed surface S gives $\oiint_S d\Omega = 4\pi$,

Total flux through the closed surface:

$$\oiint_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{4\pi\varepsilon_0} \oiint_S d\Omega.$$

But

$$\oiint_S d\Omega = 4\pi \quad \text{(as long as } S \text{ encloses the charge)},$$

hence

$$\boxed{\oiint_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{4\pi\varepsilon_0}(4\pi) = \frac{Q}{\varepsilon_0}.}$$

From the sphere to any closed surface S (as we proved this with solid angle)

$$\boxed{\oiint_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q_{\text{enc}}}{\varepsilon_0}.}$$

Case III Charge outside the sphere

Flux entering:

$$\Phi_{\text{entering}} = \frac{Q}{4\pi\varepsilon_0} \frac{1}{\rho_1^2} dS_1^{\perp}.$$

Flux leaving:

$$\Phi_{\text{leaving}} = \frac{Q}{4\pi\varepsilon_0} \frac{1}{\rho_2^2} dS_2^{\perp}.$$

$$\Rightarrow \quad \text{Total flux} = \oiint_S \mathbf{E} \cdot \hat{\mathbf{n}}\, dS = 0.$$

In summary:

$$\boxed{\oiint_S \mathbf{E} \cdot \hat{\mathbf{n}}\, dS = \begin{cases} 0, & \text{if } Q \text{ is } \textit{not} \text{ in the volume enclosed by } S, \\ \dfrac{Q}{\varepsilon_0}, & \text{if } Q \textit{ is} \text{ in the volume enclosed by } S. \end{cases}}$$

$\Rightarrow$ Maxwell's 1$^{\text{st}}$ equation (Gauss's law) in integral form.

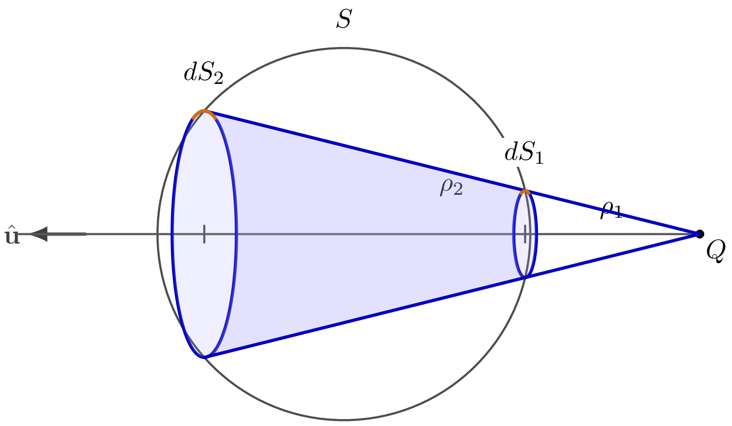


Figure 6: Case III (charge outside): the *full* cone of directions (0 to 2π) intersects the sphere in two *full* circles (rims). Distances from Q to the two intersection circles along the axis are ρ_1 and ρ_2.

From integral Gauss' law to the differential form

Starting from Gauss' law (integral form) for a closed surface $S = \partial V$:

$$\oiint_S \mathbf{E} \cdot \hat{\mathbf{n}}\, dS = \frac{Q_{\text{enc}}}{\varepsilon_0}.$$

By the divergence theorem,

$$\oiint_S \mathbf{E} \cdot \hat{\mathbf{n}}\, dS = \iiint_V (\nabla \cdot \mathbf{E})\, dV.$$

Hence,

$$\iiint_V (\nabla \cdot \mathbf{E})\, dV = \frac{Q_{\text{enc}}}{\varepsilon_0}.$$

But in reality, charge is often spread continuously in space. We describe this using the **volume charge density** $\rho_v(\mathbf{r})$:

$$\rho_v(\mathbf{r}) \equiv \lim_{\Delta V \to 0} \frac{\Delta Q}{\Delta V}, \qquad \Rightarrow \qquad dQ = \rho_v\, dV.$$

Therefore the enclosed charge in the volume V is

$$Q_{\text{enc}} = \iiint_V \rho_v\, dV.$$

Substituting into the previous relation gives

$$\iiint_V (\nabla \cdot \mathbf{E})\, dV = \frac{1}{\varepsilon_0} \iiint_V \rho_v\, dV \quad \Rightarrow \quad \iiint_V \left(\nabla \cdot \mathbf{E} - \frac{\rho_v}{\varepsilon_0} \right) dV = 0.$$

Since this holds for *any* arbitrary volume V, the integrand must be zero pointwise:

$$\boxed{\nabla \cdot \mathbf{E} = \frac{\rho_v}{\varepsilon_0}.}$$

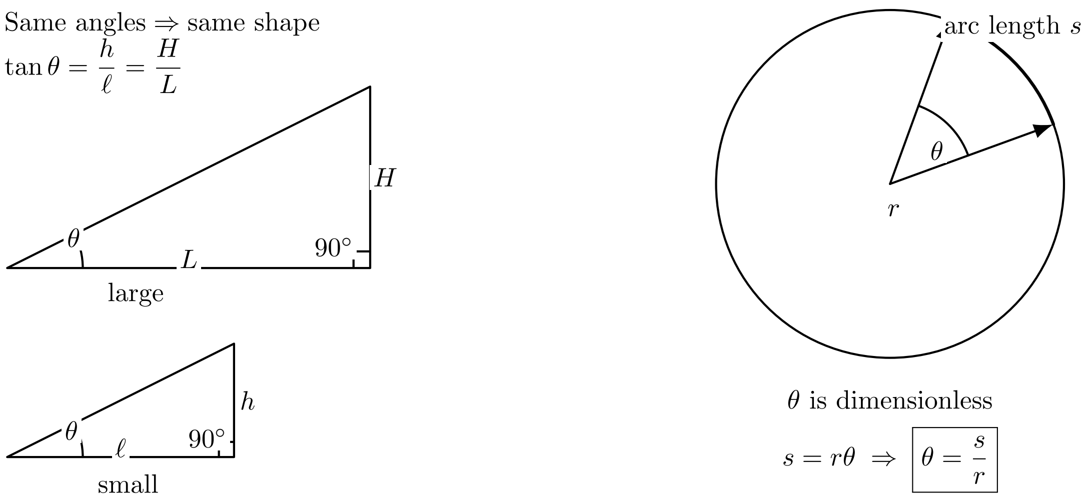
This is **Maxwell's first equation** (Gauss' law) in **differential form**.

Angle in 2D as a dimensionless "shape" quantity $\Rightarrow$ Solid angle in 3D

Core idea (2D): An angle θ characterizes the *shape* of a wedge of rays in 2D, independent of overall size. That is why **similar triangles** (same angles) represent the *same shape*.

Core idea (3D): In 3D, an "angular wedge" becomes a **cone of directions**. The dimensionless measure of the *cone's shape* (independent of size) is the **solid angle**:

$$\boxed{\Omega \;=\; \frac{\text{area cut out on a sphere}}{(\text{sphere radius})^2}} \quad\Longleftrightarrow\quad \boxed{d\Omega = \frac{dA}{r^2}}$$



(a) Same θ (and 90°) ⇒ similar triangles; angle measures shape, not size.

(b) On a circle, $\theta = s/r$: a ratio ⇒ dimensionless.

Figure 7: Angle in 2D is a dimensionless descriptor of wedge shape (independent of scale).

Takeaway: 2D angle vs. 3D solid angle (both are dimensionless)

- **2D:** θ captures wedge shape; on a circle $\theta = \frac{s}{r}$ (dimensionless).
- **3D:** Ω captures cone shape; on a sphere $\Omega = \frac{A}{r^2}$ (dimensionless).
- **Total around a point:** $\iint d\Omega = 4\pi$.

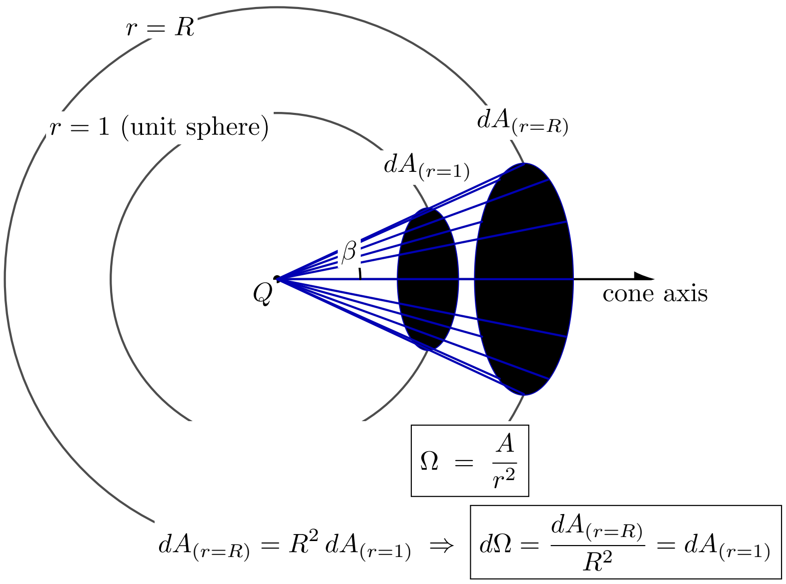


Figure 8: 3D analogue: a cone of directions cuts out a patch on a sphere. The dimensionless measure of cone shape is the solid angle $\Omega = A/r^2$. (Full cone rim shown.)

Solid angle: unit-sphere definition (core idea)

- **Set-up:** Place a sphere of radius 1 (unit sphere) centered at the observation point Q.
- **Definition:** Any surface patch (or aperture) that is "seen" from Q subtends a cone of directions. That cone cuts out a patch on the *unit sphere*. The *area* of this cut-out is the **solid angle**:

$$\boxed{d\Omega \equiv dA_{(r=1)}}$$

  where $dA_{(r=1)}$ is the area element on the unit sphere.
- **Scaling to radius R:** The same cone cuts out an area $dA_{(r=R)}$ on a sphere of radius R. Since areas scale as R^2,

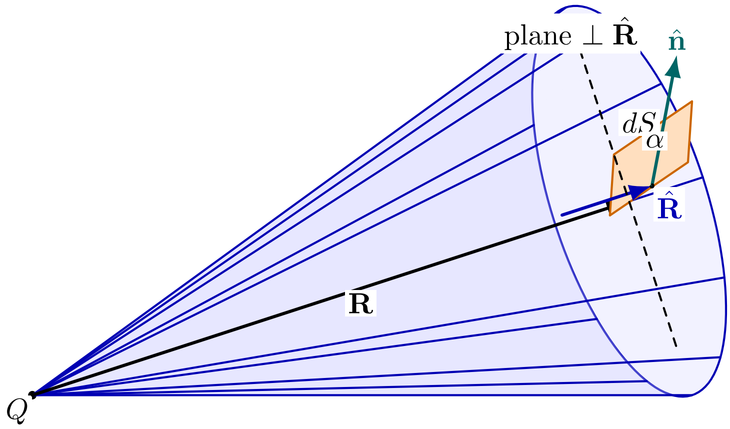
$$dA_{(r=R)} = R^2\, dA_{(r=1)} \quad \Rightarrow \quad \boxed{d\Omega = \frac{dA_{(r=R)}}{R^2}}.$$

- **Useful local formula (projection form):** For a small surface element dS located at distance R from Q, with unit normal $\hat{\mathbf{n}}$, let α be the angle between $\hat{\mathbf{n}}$ and the viewing direction $\hat{\mathbf{R}}$ (from Q to the element). Then the element contributes by its *perpendicular (projected) area* $dS_\perp = dS\cos\alpha$, giving

$$\boxed{d\Omega = \frac{dS\cos\alpha}{R^2}}.$$

- **Sanity check:** A complete closed surface around Q covers all directions once, so

$$\boxed{\iint_{\text{closed}} d\Omega = 4\pi}.$$



$$d\Omega = \frac{dS_\perp}{R^2} = \frac{dS\cos\alpha}{R^2}$$

Figure 9: Full cone-of-directions: the cone is a *complete* $0 \to 2\pi$ sweep (visible rim circle). The tilt of dS gives the projection factor $\cos\alpha$.

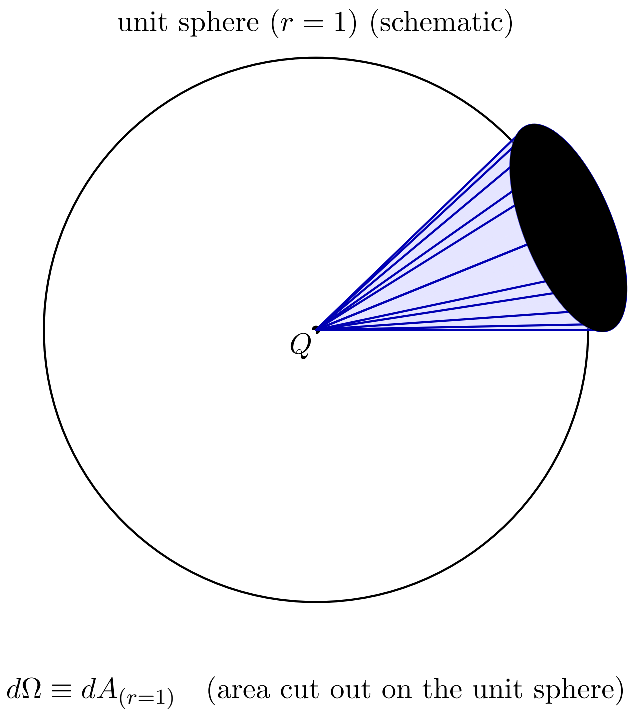


Figure 10: Unit-sphere definition with a *full* cone and a *full* rim circle. The cone cuts out an area on the unit sphere; that area is the solid angle.